QUESTION 1 (TRUE or FALSE) **[10]**

Please answer this question directly on EC. This question is also available on EC.
State whether the following statements are TRUE or FALSE.

	Statements
1.1	The three loops we use are for, while, and do-while.
1.2	The equals() method in Java returns true or false when comparing the contents of two Strings.
1.3	Multiplication and division are performed after addition and subtraction, unless parentheses are used.
1.4	In Java, variable names may begin with a number.
1.5	In Java, the == operator should always be used to compare the content of two Strings.
1.6	Omitting (leaving out) a break in a switch statement can cause multiple cases to execute.
1.7	Declaring a variable with the final keyword in Java prevents its value from being changed once a value is assigned.
1.8	A do-while loop in Java is a post-test loop because the condition is checked after the body executes once. (check answer in the memo)
1.9	Variable names must be meaningless and discreet.
1.10	In a while loop, a loop control variable is always declared inside the loop header.

QUESTION 2 (MULTIPLE CHOICE) **[10]**

Please answer this question directly on EC. This question is also available on EC.
Select the correct answer for the following questions.

2.1	What is the first time you assign a value to a variable called?
	A. Declaration
	B. Debugging
	C. Initialization
	D. Execution
2.2	Which Java statement/method/word is used to display output on the screen and then move the cursor to the next line?
	A. print
	B. println
	C. output
	D. show
2.3	Which Java keyword is in a selection structure?
	A. for
	B. switch
	C. class
	D. Declare
2.4	Which loop checks the condition before running the body?
	A. while loop
	B. do-while loop
	C. repeat loop
	D. endless loop
2.5	Which operator has the highest precedence in programming calculations?
	A. Addition +
	B. Subtraction -
	C. Multiplication *
	D. Parentheses ()
2.6	Which loop is guaranteed to run at least once?
	for loop
	while loop
	do-while loop
	repeat-until loop

2.7	Which method is used to test two Strings for equality?
	A. match()
	B. ==
	C. equals()
	D. compare()
2.8	Widening conversion is also called:
	A. Promotion
	B. explicit type conversion.
	C. Error handling
	D. Assignment
2.9	Which of the following best describes “output” in an algorithm?
	A. Data stored for later use
	B. Information displayed to the user
	C. Instructions for the computer
	D. Numbers entered by the user
2.10	Which file extension is required when saving a Java source code file?
	A. . class
	B. . java
	C. . src
	D. .txt

QUESTION 3 (SHORT ANSWERS) **[16]**

Please answer this question directly into the provided space on EC.

3.1 Identify the operator that increases the value of a variable by one in programming? (1)

3.2 Define a variable in programming. (1)

3.3 Explain the difference between declaration and initialization in Java. (2)

3.4 You're building a space exploration simulation. Declare a long variable called **distanceToMars** and set it to 225 000 000L to represent the average distance to Mars in kilometers. (3)

3.5 Distinguish when to use an if..else instead of a simple if statement? (2)

3.6 Explain why parentheses are important in order of operations. (2)

3.7 List the three main types of loops in Java and explain when to use each loop. (3)

3.8 Explain why using flags can be more effective than writing long compound conditions in a program. (2)

QUESTION 4 (SHORT PROGRAMS) [15]

4.1 Season Decoder (10)

- Download the file **SeasonDecoder.txt**, which contains a partially written Java program, and save it as a .java file (or copy the code on the next page into Notepad++).
- Complete the program by adding logic that matches numbers to seasons of the year.

The program already:

- Declares the necessary variables.
- Asks the user to enter a number between 1 and 4.

You must add the decision structure that:

- Displays the correct season name:
  - 1 → Summer
  - 2 → Autumn
  - 3 → Winter
  - 4 → Spring
- If the number is not between 1 and 4, display: **"Invalid season number."**

Do **not** change the variable names in the given code. Only complete the decision structure.

```
Enter a number (1-4): 6
Invalid season number
```

Figure 1: Output when the user enters a number outside the range (invalid input).

```
Enter a number (1-4): 1
Summer
```

Figure 2: Output when the user enters a valid number.

```
import java.util.Scanner;

public class SeasonDecoder
{
   public static void main(String[] args)
        {
       //Declarations and Input ===
       Scanner input = new Scanner(System.in);
       int seasonNumber;
           String message = "";

       System.out.print("Enter a number (1-4): ");
       seasonNumber = input.nextInt();

       //Processing
       //Your decision structure goes here(We only need the decision structure)

           //Output
           System.out.println(message);
   }
}
```

4.2 Write a count-based loop in Java that will print the numbers from 1 to 35. (5)

For each number, display whether it is divisible by 5 or not.

Only write the loop with the correct if structure and output statements. Do not write the full program. Anything other than the loop and if statement will not be marked

```
1 is not divisible by 5
2 is not divisible by 5
3 is not divisible by 5
4 is not divisible by 5
5 is divisible by 5
6 is not divisible by 5
7 is not divisible by 5
8 is not divisible by 5
9 is not divisible by 5
10 is divisible by 5
11 is not divisible by 5
12 is not divisible by 5
13 is not divisible by 5
14 is not divisible by 5
15 is divisible by 5
16 is not divisible by 5
17 is not divisible by 5
18 is not divisible by 5
19 is not divisible by 5
20 is divisible by 5
21 is not divisible by 5
22 is not divisible by 5
23 is not divisible by 5
24 is not divisible by 5
25 is divisible by 5
26 is not divisible by 5
27 is not divisible by 5
28 is not divisible by 5
29 is not divisible by 5
30 is divisible by 5
31 is not divisible by 5
32 is not divisible by 5
33 is not divisible by 5
34 is not divisible by 5
35 is divisible by 5
```

Figure 3: Output of the loop classifying numbers 1–35 as divisible by 5 or not.

QUESTION 5 (PROGRAM PLANNING) **[14]**

- **For each block, enter exactly what belongs in the block in the corresponding letter on EC.**
- **Do not type the letter, only the block content.**
- **Use the given variable names.**

Below is the structure of a flowchart, with blank spaces. Based on the sample output and the following task description, complete the flowchart by filling in the missing operations or decisions.

You are writing a program that:

- Asks the user whether they want to convert metres to centimetres. Use the variable name ***answer*** to store their response.
- If the response is 'Y' or 'y', ask for the length in metres. Use the variable name ***metres*** to store this input.
- Calculate the length in centimetres using the formula: **cm = metres * 100.**
- Display the result.
- Repeat the process until the user enters 'N' or any input other than 'Y'.
- When the program terminates display the following message: **"You are done converting lengths."**

Example output:

```
Do you want to convert metres to centimetres? (Y/N): n
You are done converting lengths.
```

Figure 4: Sample output when the user chooses N immediately (no conversion).

```
Do you want to convert metres to centimetres? (Y/N): y
Enter length in metres: 20
20.0 metres = 2000.0 centimetres.
Do you want to convert metres to centimetres? (Y/N): n
You are done converting lengths.
```

Figure 5: Sample output when the user chooses Y, enters a length, then chooses N to finish.

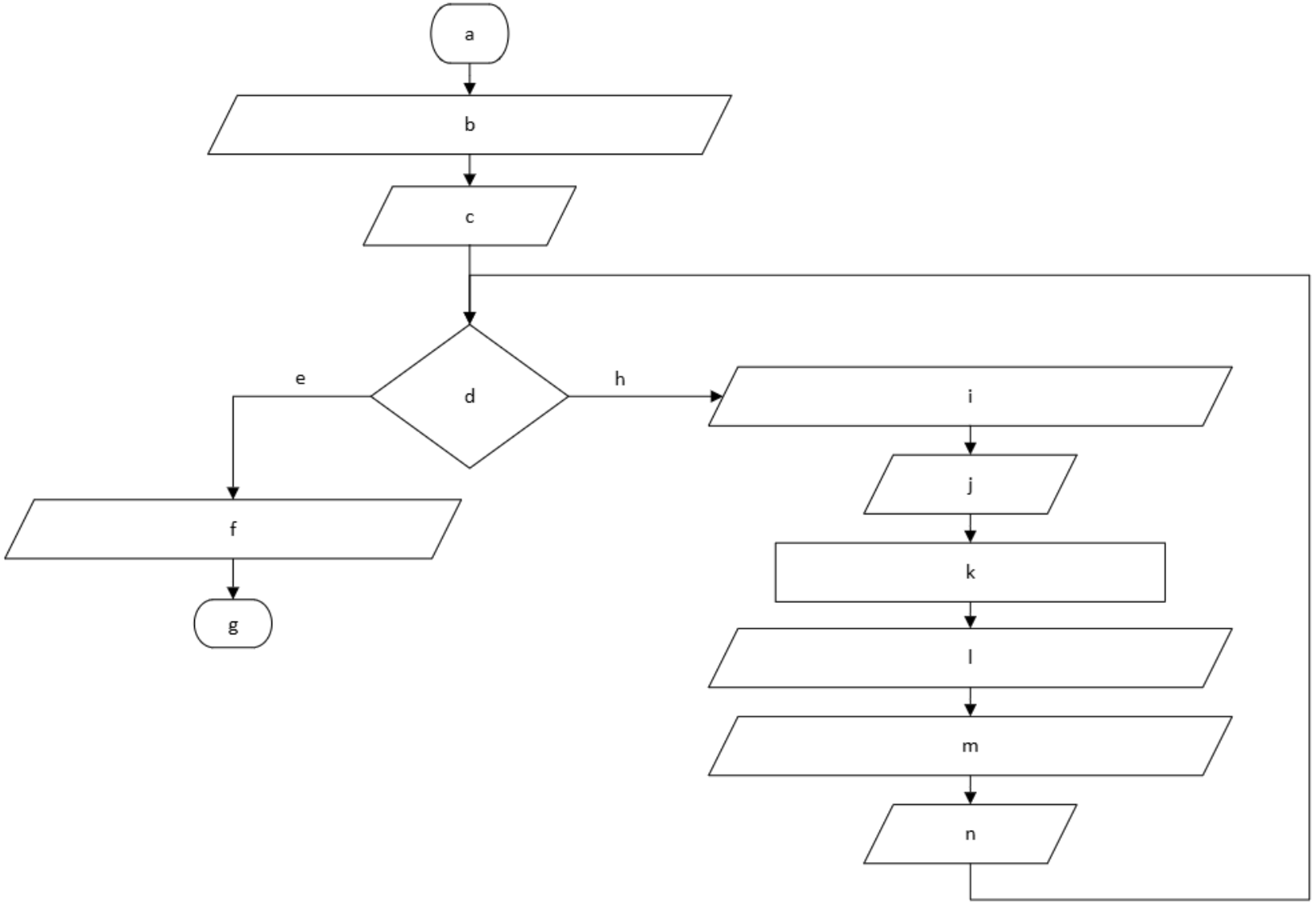


Figure 6: Flowchart structure of the metre to centimetre calculator

QUESTION 6 (DEBUGGING/WORKING WITH ERRORS) [15]

- **Download TillSlip.txt or copy from the next page.**
- **Open the file in Notepad++ and save it as TillSlip.java.**
- **Fix the logic based on the statements below.**
- **Once you are done, copy and paste it into the provided space in EC.**

The program is meant to:

- Ask how many **items** to process.
- For each item, read **price** (R) and **quantity**.
- Compute **line total = price × quantity** and add to a **grand total**.
- After the loop, compute **VAT at 15%** and display:
    - **Subtotal**, **VAT**, and **Total incl. VAT**, formatted to **two decimals** and prefixed with **"R "**.

However, the code below has **15 errors** (both syntax and logical errors).
Identify and correct them so the program compiles and works as intended.

```
Enter number of items: 2

Enter price (R) for item 1: 21.99
Enter quantity for item 1: 2

Enter price (R) for item 2: 15
Enter quantity for item 2: 6

Subtotal: R 133.98
VAT (15%): R 20.10
Total incl. VAT: R 154.08
```

Figure 7: Output of the corrected TillSlip program

```
Enter number of items: 0

Subtotal: R 0.00
VAT (15%): R 0.00
Total incl. VAT: R 0.00
```

Figure 8: Output when zero items were processed.

```
import java.util.scanner;
import java.text.DecimalFormatt;

public class TillSlip
   public static void main(String arg)
   {
                  //Declare
      int numItems
      double price qty, lineTotal;
      double grandTotal;
      double vatAmount, totalIncl;
      int i;
      Scanner input = new scanner(System.in);
      DecimalFormat df = new DecimalFormatt("0.00");

                  //Input
      System.out.print("Enter number of items: ")
      numItems = input.nextInt();

                  //Loop that runs the number of times the user enter above
      for (i = 1; i <= numitems; i++)
                  {
                        //Input the price
         System.out.print("\nEnter price (R) for item " + i + ": ");
         price = input.nextdouble();
         //Input the quantity
                        System.out.print("Enter quantity for item " + i + ": ");
         qty = input.nextDouble();

                        //Calculate the how much to pay for the products
         lineTotal = price + qty;
                        //Add products prices to the overall total
         grandTotal =+ lineTotal;
      }

                  //Calculate VAT
      vatAmount = grandTotal * 0.015;
                  //The total after VAT
      totalIncl = grandTotal + vatAmount;

                  //Display the results
      System.out.println("\nSubtotal: R " + df.formatted(grandTotal));
      System.out.println("VAT (15%): R " + df.format(vatAmount));
      System.out.println("Total incl. VAT: R " + df.format(totalIncl));
   } // end main
} // end class
```

QUESTION 7 (PROBLEM SOLVING) [20]

• Create a java program in Notepad++ based on the statements below.

• Once you are done, copy and paste it into the provided space in EC.

The Soshanguve Teen eSports League (Ward 18) needs a tool to check if applicants qualify for the league's teens-only registration. Your task is to write a Java program called: **TeenEsportsEligibility**

The program must do the following:

- Display a clearly formatted welcome message that mentions "Soshanguve Teen eSports League – Ward 18" (follow example output style).
- Allow the user to enter the applicant's full name. Ensure the name is at least 2 characters long. If it's shorter, ask again.
- Allow the user to enter the applicant's age. The age must be between 0 and 130, or the program must ask again. (Ensure to clear the buffer after, in order to avoid issues).
- If the person is a teen (between twelve and twenty), display that they are eligible to participate. Otherwise, say they are not eligible.
- After each applicant is processed, ask the user if they want to check another applicant. Accept only Y or N (case insensitive). The loop must run at least once.
- Once all applicants have been processed, display a summary of:
    - Total people checked
    - How many were eligible
    - How many were not
- Finally, display a goodbye message.

```
==============================================
 Welcome to the Soshanguve Teen eSports League
              Ward 18 Registration
==============================================
Enter applicant's full name: Lucky
Enter player's age (0 - 130): 70
Lucky is NOT eligible to participate.
Do you want to check another applicant? (Y/N): y
Enter applicant's full name: Oprah
Enter player's age (0 - 130): 13
Oprah is ELIGIBLE to participate in the Teen eSports League.
Do you want to check another applicant? (Y/N): n

=========== SUMMARY ===========
Total applicants checked : 2
Eligible applicants      : 1
Not eligible applicants  : 1
================================
Thank you for using the Soshanguve Teen eSports League - Ward 18 system. Goodbye!
```

Figure 9: Sample output of *TeenEsportsEligibility* — two applicants checked (one eligible, one not) with loop repeat and summary.

```
==============================================
 Welcome to the Soshanguve Teen eSports League
              Ward 18 Registration
==============================================
Enter applicant's full name: Musa
Enter player's age (0 - 130): 12
Musa is NOT eligible to participate.
Do you want to check another applicant? (Y/N): n

=========== SUMMARY ===========
Total applicants checked : 1
Eligible applicants      : 0
Not eligible applicants  : 1
================================
Thank you for using the Soshanguve Teen eSports League - Ward 18 system. Goodbye!
```

Figure 10: Output showing the program processes the first applicant once, even when the user chooses “N,” then prints the summary.